Statistical and Computational Models

Lecture 13: Approximating the posterior

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Roadmap

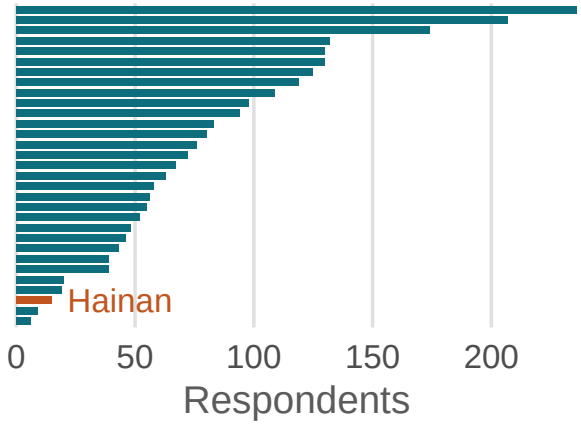
Priors and posteriors

Approximating the posterior

Trust in local government

Suppose we survey 2,500 adults across China's 31 provinces:

province	age	educ	hukou	trust
Yunnan	35-59	Secondary	Rural	1
Shanxi	18-34	College	Rural	0
Guangdong	35-59	Secondary	Urban	1
Jiangsu	18-34	College	Urban	1
Guangdong	35-59	College	Urban	0
Hainan	18-34	College	Urban	1



Y_i = 1 if respondent *i* trusts local government, 0 otherwise.

Target: π, the share of **all** Hainan adults who trust local government.

Without seeing data: what's your best guess for π, and how sure are you?

Estimating trust in Hainan

But we do have some data! Suppose for Hainan we have 15 respondents, 12 of whom said "yes". What natural estimators might we try?

HAINAN ONLY

Subset to Hainan, and find the MLE, the sample share:

$$\hat{\pi} = r/n = 12/15 = 0.80$$

$$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/15} = 0.103$$

$$\hat{\pi} \pm 1.96\,\text{se} = [0.598,\ 1.002]$$

Assumes a **random sample** of Hainan's adults. Our interval is wide because we only use limited data.

ALL PROVINCES

Pool all 2,500 respondents, and find the MLE:

$$\hat{\pi} = 1{,}449/2{,}500 = 0.58$$

$$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/2{,}500} = 0.010$$

$$\hat{\pi} \pm 1.96\,\text{se} = [0.560,\ 0.599]$$

Assumes Hainan is **like the others**. Precise, but Hainan is 0.6% of the data, so it may miss Hainan.

Bias and variance

Re-run the survey 1,000 times, as random samples of the same size, and average. We know π, since we simulated it.

Estimator	π	One draw	Average	Spread (sd)
No pooling				
Hainan (n = 15)	0.67	0.80	0.68	0.12
Guangdong (n = 236)	0.56	0.50	0.56	0.03
Complete pooling				
Hainan (n = 2,500)	0.67	0.65	0.65	0.01

No pooling, Hainan vs. Guangdong: both are right on average, but with 15 respondents any one survey can land far off. High **variance**: alone, a small province **overfits**.

Complete pooling: almost no spread, but always off, by Hainan's gap from the national share. High **bias**: pooling **underfits**.

Bayesian inference

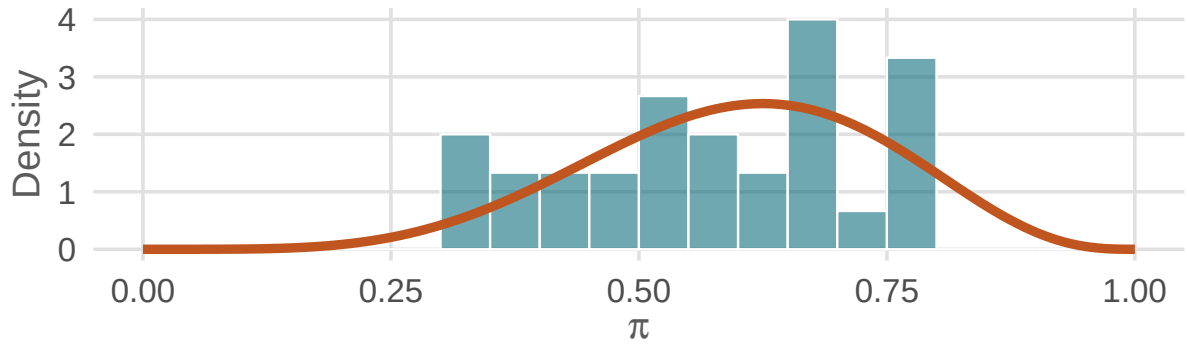
Maximum likelihood assumed a probability model for $\boldsymbol{Y}$ only. Suppose we **also** assume a distribution for the parameter, before we see the data.

1. **Assume** a model for the data (here, as before, $Y_i \sim \text{Bernoulli}(\pi)$), and a **prior** $p(\pi)$: how plausible each value of π is.
2. **Guess** values of the parameter, and weight each by its prior, times how likely the data would be had it been the true value.
3. **Learn** a distribution for the parameter given the data, the **posterior** $p(\pi \mid \mathbf{Y})$, not a single number.

That is, π is still fixed and unknown. The prior and the posterior describe **our uncertainty** about it.

A prior for Hainan

Your guesses, and how sure you were of them, are a prior. Another source is the other 30 provinces' sample shares (bars): centre 0.58, sd 0.15.



To turn the bars into a prior, we need a distribution on $[0, 1]$. A typical choice is the **Beta** distribution, whose parameters a and b act like ones and zeros already seen:

$$\pi \sim \text{Beta}(a, b), \qquad \mathbb{E}[\pi] = \frac{a}{a + b}$$

Choose a and b to match the bars: Beta(6, 4) has centre 0.60, sd 0.15 (curve). So it is **worth 10 respondents**: as if we had already asked 10 Hainan adults, and 6 said yes. Against Hainan's 15, that is a lot; against Guangdong's 236, very little.

Bayes' rule

How do we combine the prior with the data? **Bayes' rule**:

$$\underbrace{p(\pi \mid \mathbf{Y})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{Y} \mid \pi)}^{\text{likelihood, } L(\pi;\mathbf{Y})} \overbrace{p(\pi)}^{\text{prior}}}{\underbrace{p(\mathbf{Y})}_{\text{normalising constant}}}$$

$$p(\pi \mid \mathbf{Y}) \propto L(\pi;\mathbf{Y})\, p(\pi)$$

Why can we drop $p(\mathbf{Y})$?

It does not depend on π: it only makes the posterior integrate to one.

Guessing π, with a prior

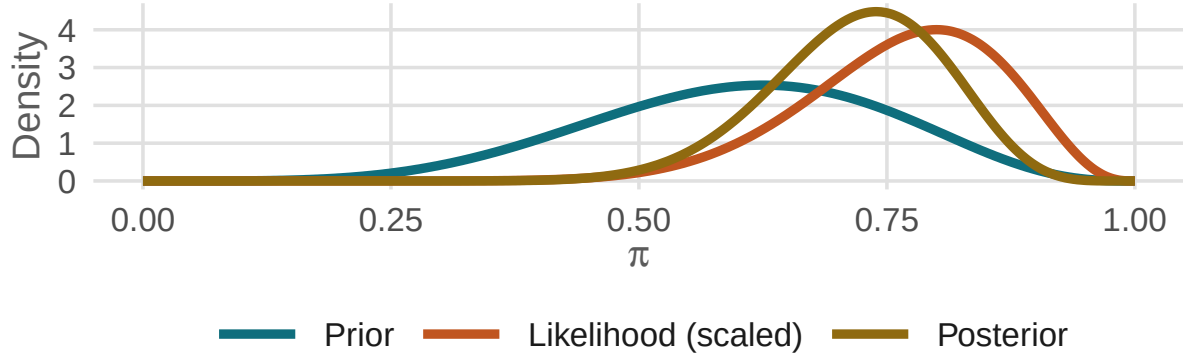
Suppose π could only be one of five values. Our Beta(6, 4) prior at each, rescaled to sum to one:

π	Prior	$L(\pi; \mathbf{Y})$	Prior $\times$ L	Posterior
0.5	0.23	3.1e-05	7.2e-06	0.03
0.6	0.30	1.4e-04	4.2e-05	0.17
0.7	0.27	3.7e-04	1.0e-04	0.41
0.8	0.16	5.5e-04	8.7e-05	0.35
0.9	0.04	2.8e-04	1.0e-05	0.04
Sum	1.00		2.5e-04	1.00

The posterior peaks at 0.7, not 0.8 (the MLE): 15 respondents are not enough to overrule a prior worth 10.

Unlike the MLE, we learn how **plausible** each value is, not just which is best.

The Beta-Binomial



Now let π take **any** value in $[0, 1]$, not just five. Hainan's prior, Beta(6, 4), becomes a curve.

Add the likelihood, rescaled to area one. Where will the posterior sit?

Between the two, nearer the likelihood: 15 respondents against a prior worth 10.

With a Bernoulli likelihood, a Beta prior gives a Beta posterior: the prior is **conjugate**.

$$\pi \mid \mathbf{Y} \sim \text{Beta}(a + r,\ b + n - r) = \text{Beta}(18, 7)$$

The prior and the sample size

Posterior means under three priors, for Hainan and for Guangdong:

Prior	Worth	Prior sd	Hainan (n = 15)	Guangdong (n = 236)
None (MLE)			0.800	0.466
Beta(1, 1)	2	0.29	0.765	0.466
Beta(6, 4)	10	0.15	0.720	0.472
Beta(12, 8)	20	0.11	0.686	0.477

The priors move Hainan up to 11 points from the MLE, but Guangdong only 1. Why?

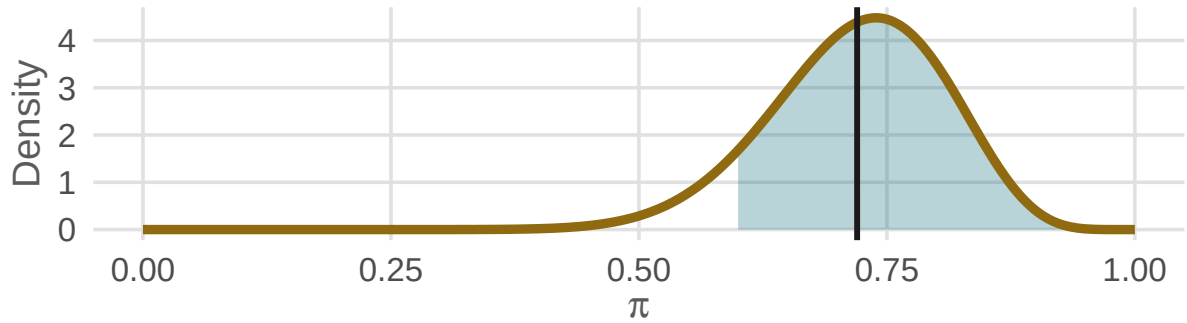
$$\mathbb{E}[\pi \mid \mathbf{Y}] = \underbrace{\frac{a+b}{a+b+n}}_{\text{weight on prior}} \cdot \frac{a}{a+b} + \underbrace{\frac{n}{a+b+n}}_{\text{weight on data}} \cdot \frac{r}{n}$$

Read: a prior worth more is **spikier**, it carries more information.

But the prior is an **assumption**, and it can be wrong!

Summarising the posterior

The posterior is a distribution, which we can **summarise**:



Point estimate: the posterior mean, 0.72 (solid line).

Credible interval: the middle 95% of the posterior, [0.53, 0.87].

Read: given our model and prior, there is a 95% probability that π lies in it. Different to a confidence interval, whose 95% describes the procedure, over **repeated samples**.

Probabilities: the area above 0.6 (shaded), $\mathbb{P}(\pi > 0.6 \mid \mathbf{Y}) = 0.90$, the chance that over 60% of Hainan's adults trust.

Roadmap

Priors and posteriors

Approximating the posterior

When the posterior has no formula

For Hainan, the posterior had a formula: Beta(18, 7). That is because its normalising constant has one:

$$p(\mathbf{Y}) = \int_0^1 L(\pi; \mathbf{Y})\, p(\pi)\, d\pi$$

For most models, it has none. For a logit, it is an integral over every coefficient at once:

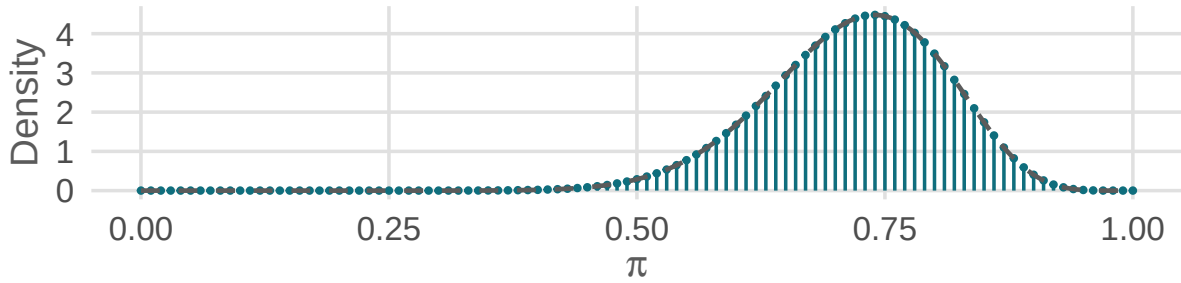
$$p(\mathbf{Y}) = \int L(\beta; \mathbf{Y}, \boldsymbol{X})\, p(\beta)\, d\beta$$

Then we must **approximate** the posterior.

Grid approximation

We have already tried an approximation: a rough **grid** of five guesses.

With many finer guesses, again dividing prior times likelihood by its sum:



The grid's mean, 0.720, matches the exact posterior's, Beta(18, 7) (dashed).

But grids grow fast: 100 points for one coefficient, 100×100 for two, etc.

Much of the grid falls where the posterior is near zero – can we spend our effort more efficiently?

Coefficients	Evaluations	Time
1	100	under a second
2	10,000	under a second
4	10^8	1.1 hours
7	10^{14}	127 years

Moving through the posterior

Let's explore Hainan's posterior: at step t, we are at $\pi^{(t)}$. Where next?

An optimiser only moves uphill, to the peak. To learn a **distribution**, we must sometimes move **downhill**: uphill always, downhill sometimes, less often the bigger the drop.

1. Start somewhere: $\pi^{(0)}$
2. Propose a random step: $\pi^* = \pi^{(t)} + \varepsilon, \quad \varepsilon \sim N(0, s^2)$
3. Compare prior times likelihood: ratio $= \dfrac{L(\pi^*; \mathbf{Y})\, p(\pi^*)}{L(\pi^{(t)}; \mathbf{Y})\, p(\pi^{(t)})}$
4. $\pi^{(t+1)} = \pi^*$ with probability $\min(1, \text{ratio})$; otherwise $\pi^{(t+1)} = \pi^{(t)}$. Repeat.

Note: $p(\mathbf{Y})$ **cancels** in the ratio, so we never need it. This is the **Metropolis** algorithm.

A Metropolis chain for Hainan

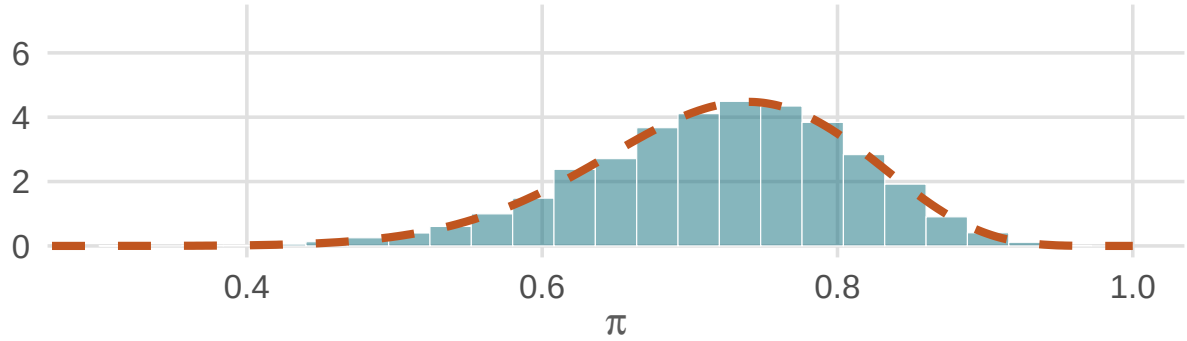
Back to Hainan: prior Beta(6, 4), steps from $N(0, 0.1^2)$, a start at 0.5. The first 1,000 draws are usually uninformative about the posterior, so we drop them ('burning in'). The next six, moving if $u <$ ratio, with u uniform on $[0, 1]$:

t	$\pi^{(t)}$	π^*	ratio	u	Move?	$\pi^{(t+1)}$
1,001	0.586	0.560	0.667	0.485	yes	0.560
1,002	0.560	0.480	0.199	0.302	no	0.560
1,003	0.560	0.431	0.055	0.605	no	0.560
1,004	0.560	0.424	0.044	0.158	no	0.560
1,005	0.560	0.671	3.798	0.812	yes	0.671
1,006	0.671	0.701	1.186	0.205	yes	0.701

1,006: uphill (ratio above 1), so we always move.

A Metropolis chain for Hainan

Keep going, and collect the draws:



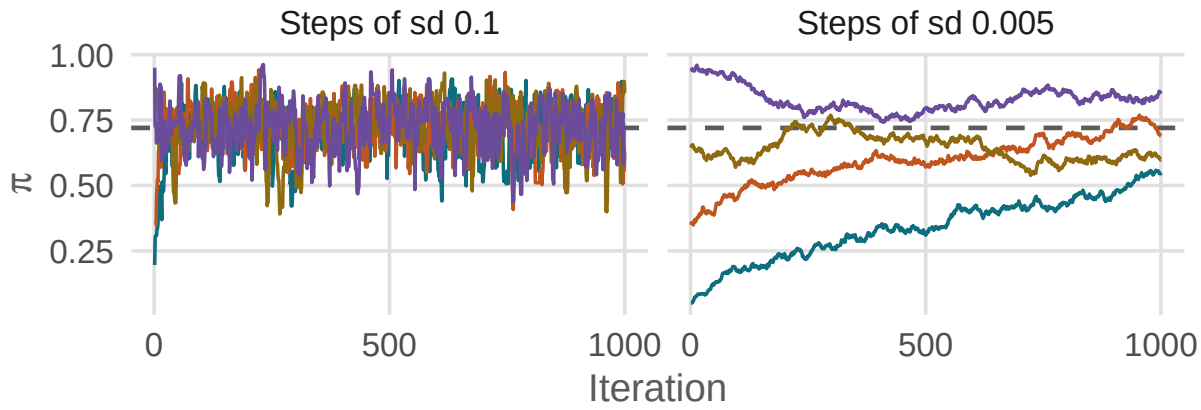
Bars: all 19,000 draws (after burning the first 1,000). Dashed: the exact Beta(18, 7).

Run long enough, the draws settle on the posterior.

A rejected step records the current value **again**, so the chain spends more time where the posterior is higher.

Each draw depends only on the one before: a **Markov chain**. We summarise the posterior with the draws: **Monte Carlo**. Together, **MCMC**.

Checking convergence



Four chains, four starts: a **trace plot** of each chain's draws.

Left: the chains meet fast, then mix like a fuzzy caterpillar.

Right: with tiny steps, they still disagree after 1,000 iterations. They have not **converged**.

Check with **R-hat**, near 1 when the chains agree (1.00 left, 3.86 right). If they agree, **combine** their draws.

References

- Efron, B. & Hastie, T. (2016) *Computer Age Statistical Inference* – **ch. 3**.
- Johnson, A. A., Ott, M. Q. & Dogucu, M. (2022) *Bayes Rules! An Introduction to Applied Bayesian Modeling* – **ch. 6**; chs. 2–5 and 7 as background. bayesrulesbook.com.